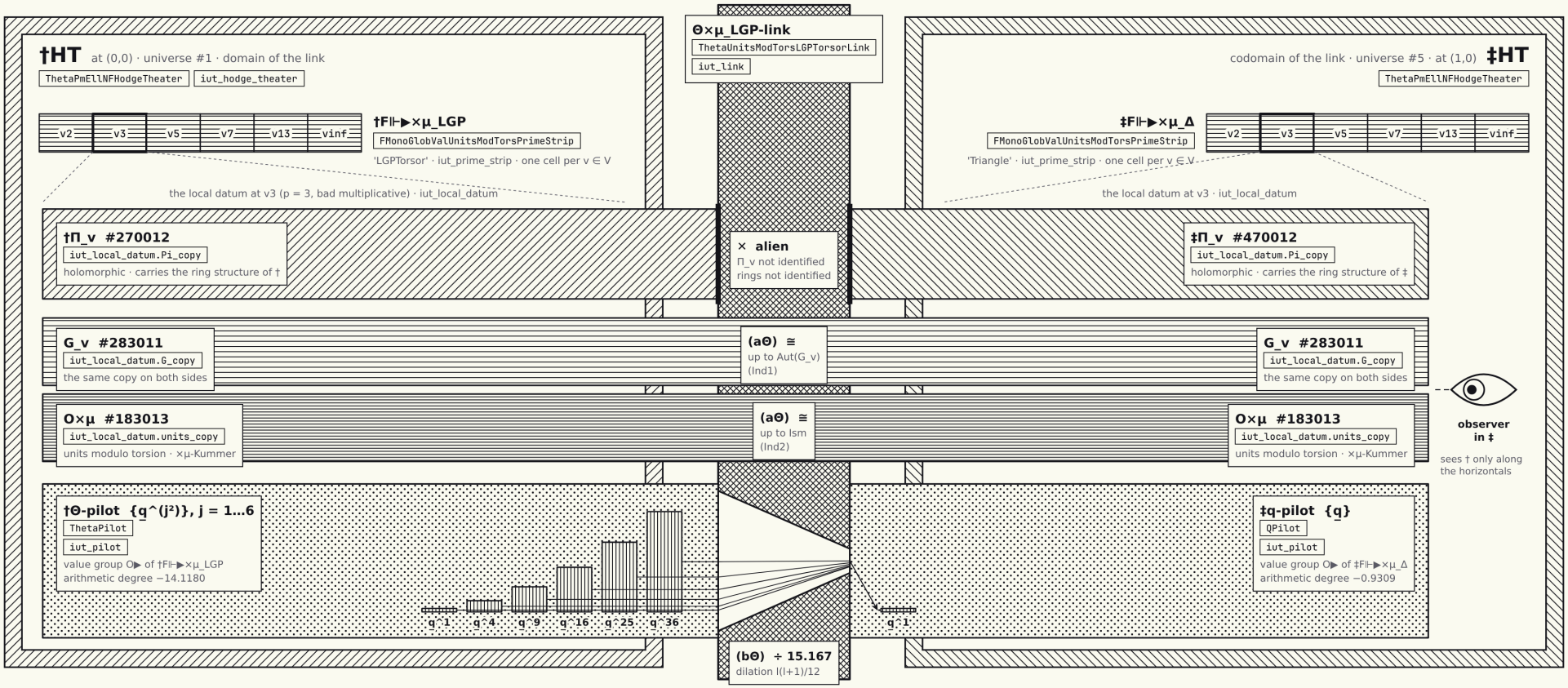
PLATE 1 / 6

Two theaters and the wall between them

The Θ×μ_LGP-link glues †HT to ‡HT along one prime-strip only. An observer in ‡ looks back at †: the horizontal lines run straight through the wall, the diagonals of † stop at it, and the dots come out rescaled.

```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l* = 6   l± = 7   |V| = 6
```



ring structure of † — one universe · (0,0) · #1

ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

mono-analytic — horizontally coric: G_v ↷ O×μ

bi-coric — D⊢ = {G_v}, shared by all

value group — O▶, pilot objects

indeterminate — up to (Ind1) (Ind2) (Ind3)

species — outline only: a type, not a set

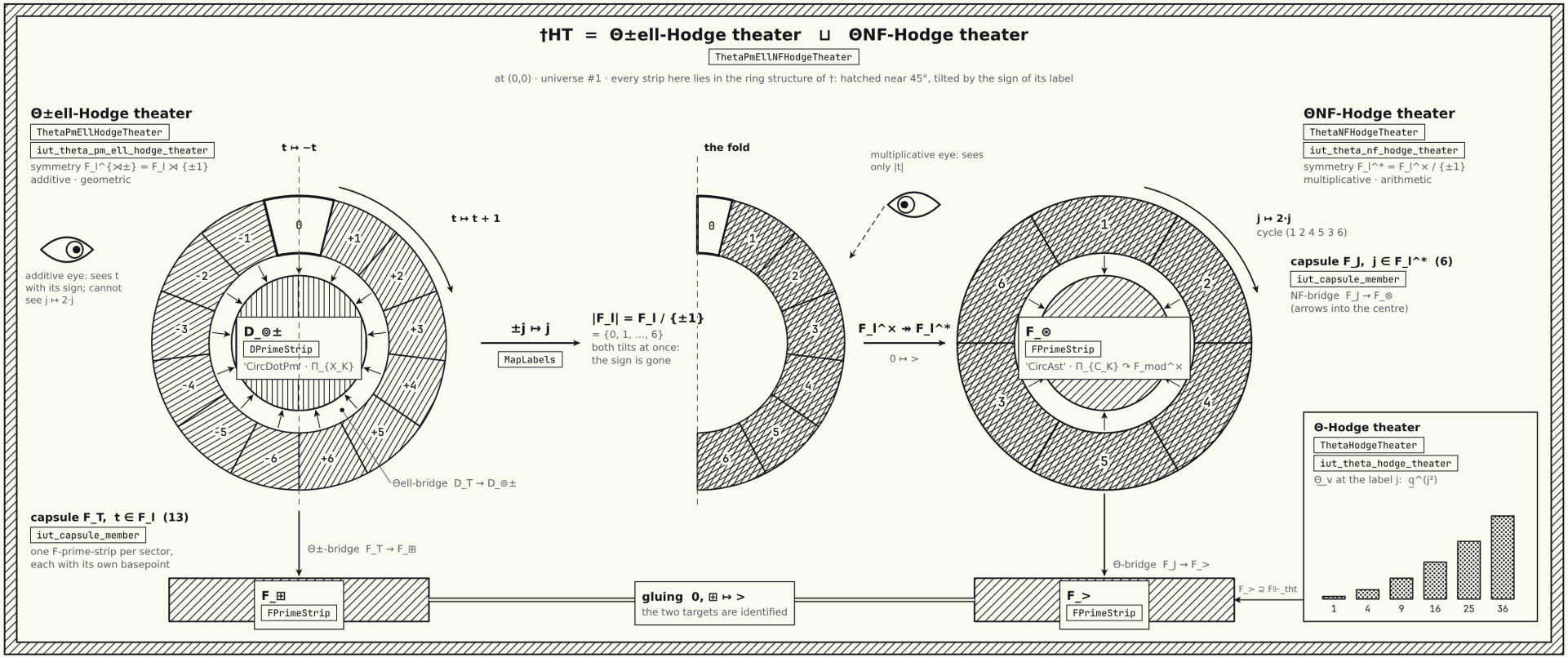
Every chip names a data type of the toolkit (species ids of `iut species`, C structs of src/iut.h). All numbers are read from `iut walk`.

scripts/infographic/hatched.js

Inside one theater: the additive eye and the multiplicative eye

One Θ±ell NF-Hodge theater is two species glued along their labels. The Θ±ell side sees a label t with its sign and turns the labels by addition; the ΘNF side sees only j = |t| and turns them by multiplication. Neither symmetry is visible from the other side.

```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l* = 6   l± = 7   |V| = 6
```



ring structure of † — one universe · (0,0) · #1

ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

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bi-coric — D⊩ = {G_v}, shared by all

value group — O▶, pilot objects

indeterminate — up to (Ind1) (Ind2) (Ind3)

species — outline only: a type, not a set

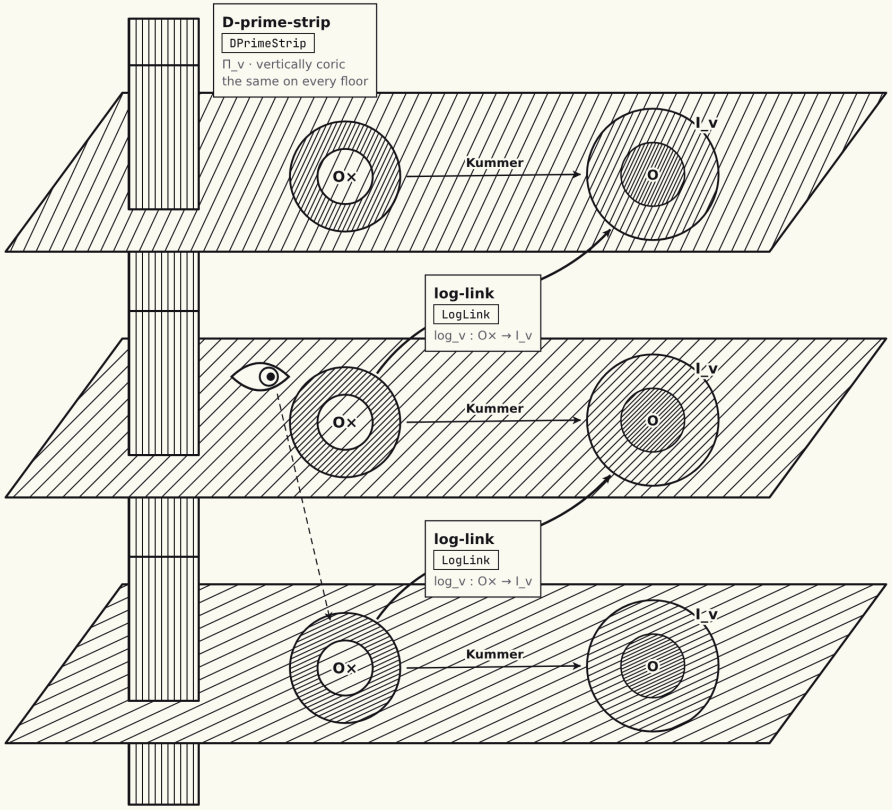
Every chip names a data type of the toolkit (species ids of `iut species`, C structs of src/iut.h). All numbers are read from `iut walk`.

scripts/infographic/hatched.js

Looking up a column: the log-link

A column of the lattice is a stack of floors, each with its own ring structure. The log-link carries the units of one floor into the additive log-shell of the floor above, so ⊠ below becomes ⊞ above. Only the étale-like pillar Π_v stands unchanged through every floor.

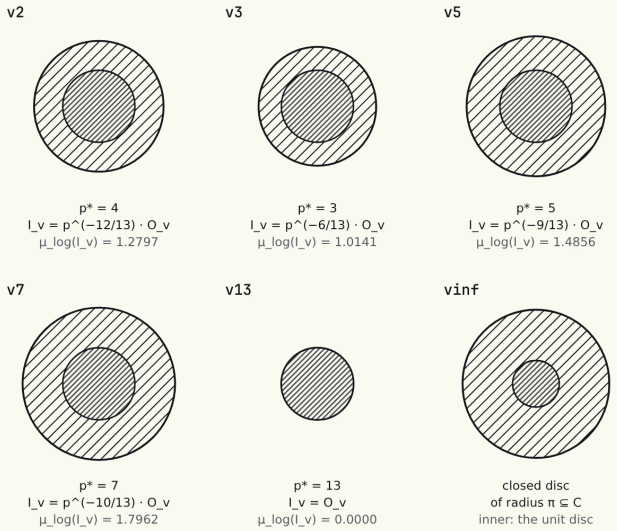
```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l* = 6   l± = 7   |V| = 6
```



The log-shells of †HT, one per v ∈ V

`LogShell` `iut_log_shell` `LogShellConstruction`

l_v = (p*)⁻¹ · log(O×_{K_v}) ⊇ O_v inner disc: O_v outer disc: l_v, radius grows with μ_log



These are the Frobenius-like, holomorphic log-shells of the floor (0,0): the codomain of the log-link from (0,−1). At a valuation of good reduction that is unramified the shell is O_v itself.

ring structure of † — one universe · (0,0) · #1

ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

mono-analytic — horizontally coric: G_v ↷ O×μ

bi-coric — D⊢ = {G_v}, shared by all

value group — O▶, pilot objects

indeterminate — up to (Ind1) (Ind2) (Ind3)

species — outline only: a type, not a set

Every chip names a data type of the toolkit (species ids of `iut species`, C structs of src/iut.h). All numbers are read from `iut walk`.

PLATE 4 / 6

The étale picture: looking through the core

Forget the Frobenius-like part of every theater and what remains can be laid out as spokes around one shared core, the D⊩-prime-strip. The only line of sight from ‡ to † runs down a spoke, through the core and up another spoke, and what arrives is known only up to three indeterminacies.

```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l* = 6   l± = 7   |V| = 6
```

outer band: the Frobenius-like part

`ThetaPmEllNFHodgeTheater`

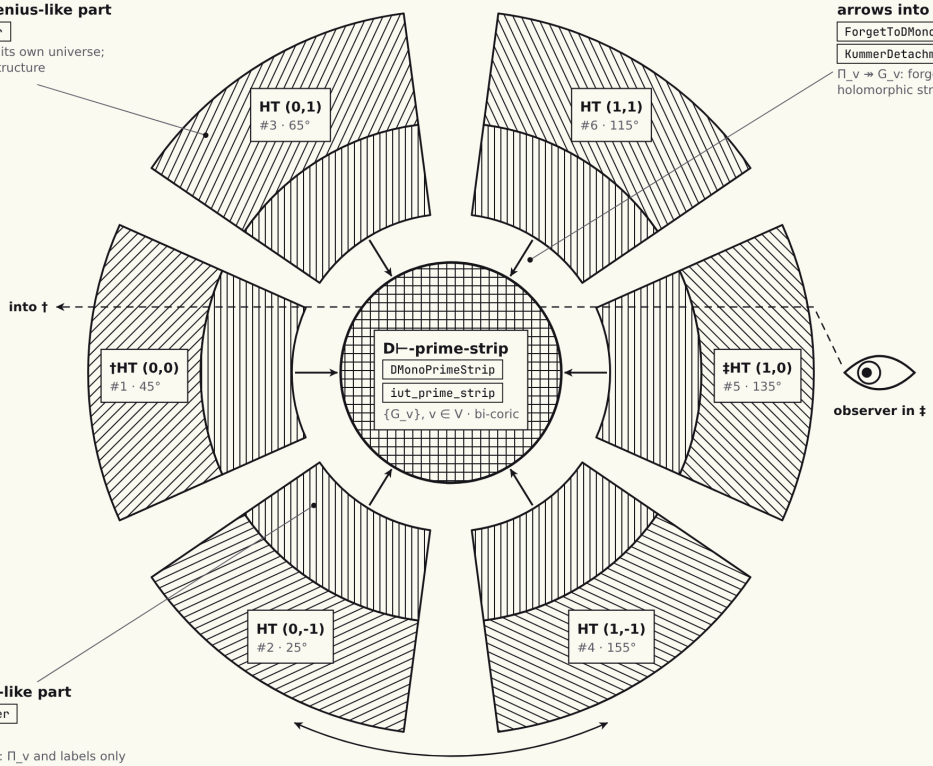
hatched in the ring angle of its own universe; no two bands share a ring structure

arrows into the core

`ForgetToDMono`

`KummerDetachment`

Π_v ↠ G_v: forget the holomorphic structure



inner band: the étale-like part

`DThetaPmEllNFHodgeTheater`

`ForgetToD`

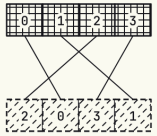
a D-Θ±ell NF-Hodge theater: Π_v and labels only

spokes may be permuted freely

the core does not know which is which

What the observer in ‡ can rebuild of †

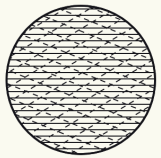
`iut_indeterminacies` `MultiradialRepresentation`



(Ind1) which is which

`iut_indeterminacies.ind1`

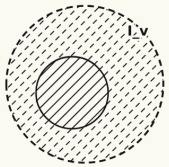
automorphisms of the processions of D⊩-prime-strips: permutations of the labels in S±_(j+1), up to (j+1)!, and Aut(G_v)



(Ind2) turned by how much

`iut_indeterminacies.ind2`

the units O×μ are shared only up to an isometry (Ism): the horizontal lines arrive turned by an unknown amount



(Ind3) no more than this

`iut_indeterminacies.ind3`

upper semi-compatibility of the log-Kummer correspondence: the region is only contained in the log-shell, an upper bound

ring structure of † — one universe · (0,0) · #1

ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

mono-analytic — horizontally coric: G_v ↷ O×μ

bi-coric — D⊩ = {G_v}, shared by all

value group — O▶, pilot objects

indeterminate — up to (Ind1) (Ind2) (Ind3)

species — outline only: a type, not a set

Every chip names a data type of the toolkit (species ids of `iut species`, C structs of src/iut.h). All numbers are read from `iut walk`.

scripts/infographic/hatched.js

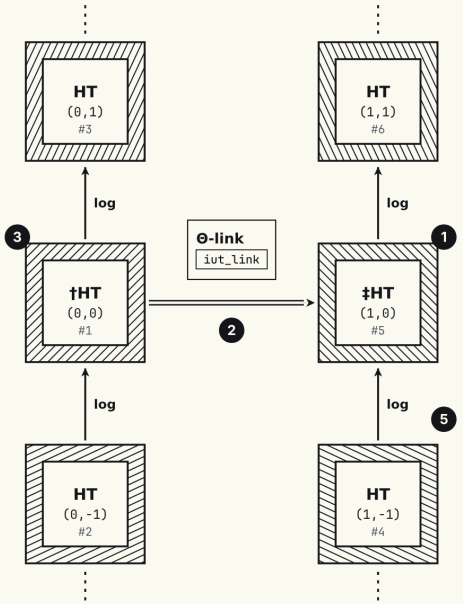
The whole process: one pilot seen from the other side

The infinite H, step by step. The Θ-pilot of † is the q-pilot of ‡ as far as the link can tell. Rebuilt inside one ring structure from the shared core, it becomes a region with fuzzy edges, and comparing log-volumes gives the inequality of Cor 3.12.

```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l* = 6   l± = 7   |V| = 6
```

The log-theta-lattice

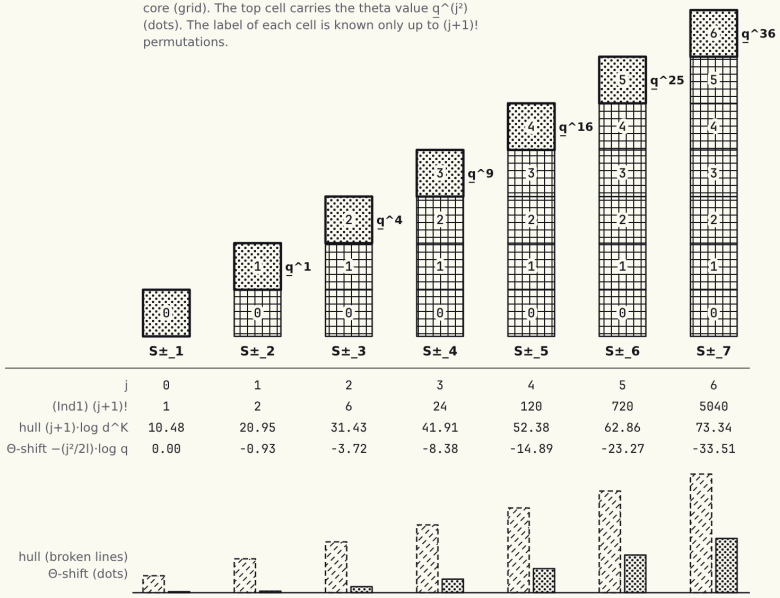
`iut_log_theta_lattice`



4 The procession: tensor packets of log-shells

`TensorPacket` `iut_procession_step` `TensorPacketConstruction`

Column j is the container S±_(j+1) = {0, …, j}: j + 1 copies of the mono-analytic log-shell, rebuilt from the core (grid). The top cell carries the theta value q̲^(j²) (dots). The label of each cell is known only up to (j+1)! permutations.

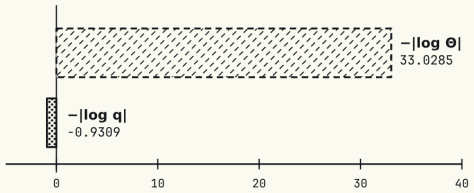


	S±_1	S±_2	S±_3	S±_4	S±_5	S±_6	S±_7
j	0	1	2	3	4	5	6
(Ind1) (j+1)!	1	2	6	24	120	720	5040
hull (j+1)·log d^K	10.48	20.95	31.43	41.91	52.38	62.86	73.34
Θ-shift −(j²/2l)·log q	0.00	-0.93	-3.72	-8.38	-14.89	-23.27	-33.51

hull (broken lines)
Θ-shift (dots)

6 The comparison of log-volumes

`MultiradialRepresentation` `iut_multiradial_rep` `MultiradialAlgorithm`



log-volume, procession-normalized

7 −|log Θ| ≥ −|log q|

`33.0285 ≥ -0.9309 holds for this data`

leading-term heuristic of the toolkit: hull enlargement 47.1465, Σ_j [(j+1)·log d^K − (j²/2l)·log q] = 198.1713, divided by l* = 6

`C_0 = 1.87999e+22 ≥ -1 ✓ ([IUTchIV] Thm 1.10)`

The observer stays inside the ring structure of one theater the whole time. The q-pilot is measured there directly (dots). The Θ-pilot is rebuilt there from the shared core, so it is known only as a region (broken lines): its log-volume is an upper bound.

Whether step 6 is a valid comparison is the disputed point of the theory ([IUTchIII] Cor 3.12). The plate draws what the toolkit computes, not a verdict.

1 ‡q-pilot

`QPilot`

log-volume −0.9309 in ‡

2 †Θ-pilot ↦ ‡q-pilot

`ThetaUnitsModTorsLGPTorsorLink`

only Fl⊩▶×μ crosses

3 Kummer detachment, log-shells

`KummerDetachment`

up and down the column of †

ring structure of † — one universe · (0,0) · #1

ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

mono-analytic — horizontally coric: G_v ↷ O×μ

bi-coric — D⊩ = {G_v}, shared by all

value group — O▶, pilot objects

indeterminate — up to (Ind1) (Ind2) (Ind3)

species — outline only: a type, not a set

Every chip names a data type of the toolkit (species ids of `iut species`, C structs of src/iut.h). All numbers are read from `iut walk`.

scripts/infographic/hatched.js

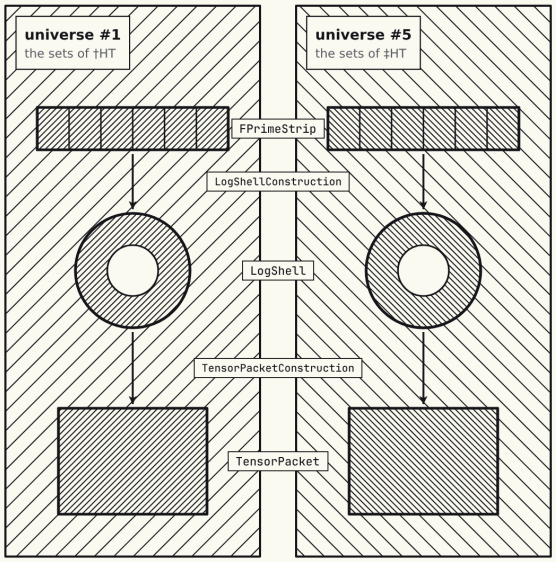
Species and mutations: what can be carried between universes

A set lives in one universe and cannot be moved. A species is only a rule for building something, so it can be run in any universe; a mutation is a rule that turns one species into another. Below, an outline is a species, and the hatch that fills it is the universe it happens to be run in.

```
example   abc(1,2400,2401) l=13
curve     Frey y^2=x(x-1)(x+2400)
l = 13   l★ = 6   l± = 7   |V| = 6
```

Two universes, one species

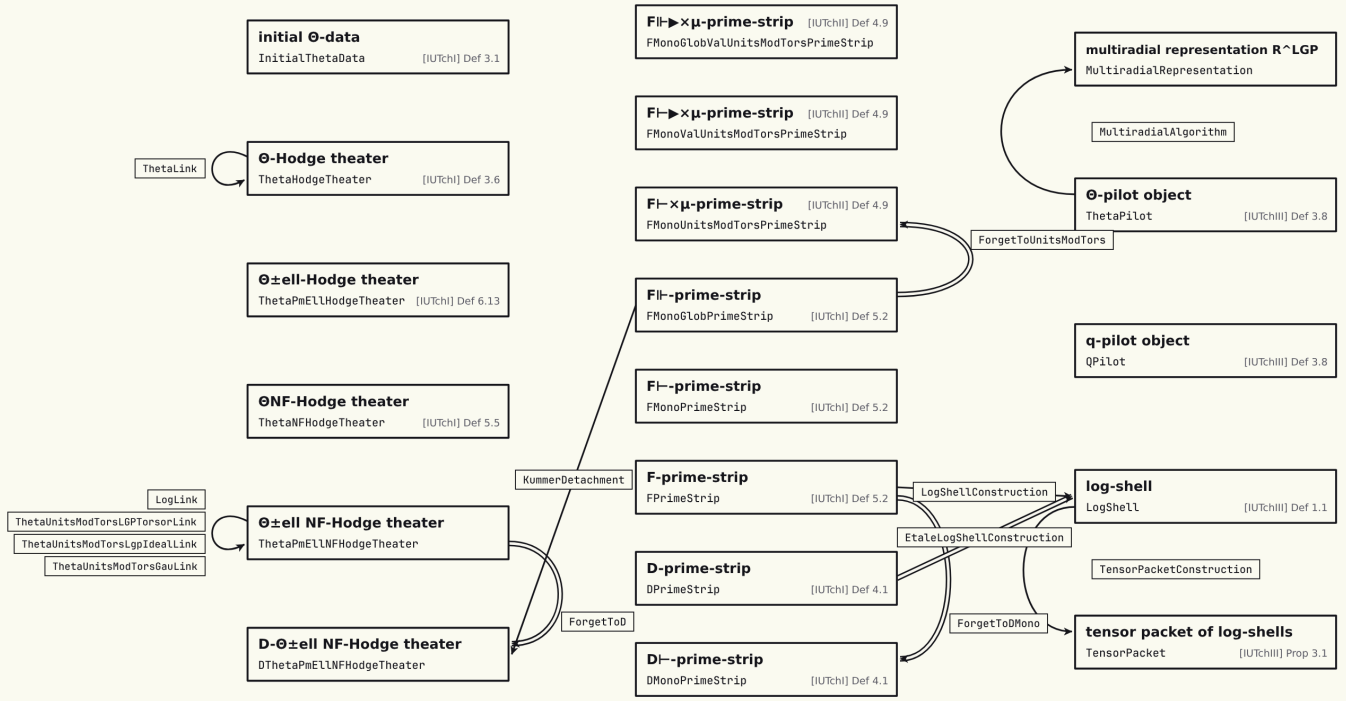
`iut_species` `iut_mutation`



Same outlines, same arrows, different lines inside. The outlines and arrows are all that can be compared between † and ‡.

The 19 species and 13 mutations of the toolkit

double arrow: a core mutation, the same seen from every universe · loop: a mutation from a species to itself



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ring structure of ‡ — another universe · (1,0) · #5

étale-like — vertically coric: Π_v, D

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